

Architecture and Design Specification

Project: Electronics Store Inventory Management System

Phase 1 — Architecture & Design

Item	Details
Version	1.0
Team	H9
Technology Stack	Python (Django)
Status	Draft / For Review

Team Members

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Part of the Team H9 Phase 1 document set: 01 SRS | 02 Test Plan | 03 Test Cases | 04 Architecture & Design.

Section and requirement numbers follow the master SRS so cross-references (RTM, VFR, DS-xx, TC-xx) remain valid across the four documents.

2. Overall Description (Design Extract)

2.4 Operating Environment

Client: modern web browser such as Chrome, Edge or Firefox. Server/application: Python runtime with Django on Windows, Linux or macOS. Database: a relational database supported by Django, such as SQLite for development or PostgreSQL/MySQL for larger deployment. Development tools may include VS Code, Git and a browser-based test environment.

Technology stack: Python (Django), Django ORM, relational database supported by Django, HTML/CSS/JavaScript client.

Design constraint (ESIM-NF-006): The Django application shall separate presentation, business logic, authentication/authorization and data-access responsibilities into identifiable modules.

7. System Models and Diagrams

7.1 UML Use-Case Diagrams

7.2 Actors and Actor-Wise Use Case Grouping

The system identifies 4 actors / external roles for Phase 1: Inventory Administrator, Store Staff, Store Manager and QA / Evaluator. The first three are operational system actors shown in the use-case diagrams. QA / Evaluator is included as an external evaluation role because the SRS explicitly assigns this role responsibility for validating the implementation against the requirements and planned acceptance tests.

Actor 1 — Inventory Administrator

Responsibilities: maintains inventory, performs permitted inventory-changing operations and manages permitted users/roles.

Grouped use cases: Login; Add Component; View Inventory; Search / Filter; Update Component; Delete Component; Check Availability; View Dashboard; Manage Users; Record Stock Change; Generate Inventory Reports.

Actor 2 — Store Staff

Responsibilities: checks inventory, availability and reports and performs only operations permitted by the assigned role.

Grouped use cases: Login; View Inventory; Search / Filter; Check Availability; View Dashboard; permitted inventory operations.

Actor 3 — Store Manager

Responsibilities: monitors inventory through dashboard, availability, low-stock information and reports.

Grouped use cases: Login; View Dashboard; Generate Report; View Availability; View Low-Stock Items; Filter Report; Export Report; Review Stock Trends; Set Reorder Level; Acknowledge Low Stock.

Actor 4 — QA / Evaluator

Responsibilities: validates the implementation against the SRS, RTM, acceptance criteria and planned security tests.

Grouped validation activities: requirements verification; functional acceptance testing; non-functional validation; security validation; penetration-testing planning/verification during the implementation and testing phase.

Actor Classification Note

The QA / Evaluator role is documented as an external evaluation role and is not treated as a business actor inside the inventory-management system boundary. The database and Django framework are internal implementation components and are not actors.

The system boundary represents the Electronics Store Inventory Management System. Only external participants are modeled as actors. The database and Django framework are internal implementation components and are not treated as business actors.

Diagram 1: Inventory & Component Management

Actors: Inventory Administrator and Store Staff. Main use cases include login, add component, view inventory, update component, delete component, search/filter, check availability, view dashboard, manage users and record stock changes.

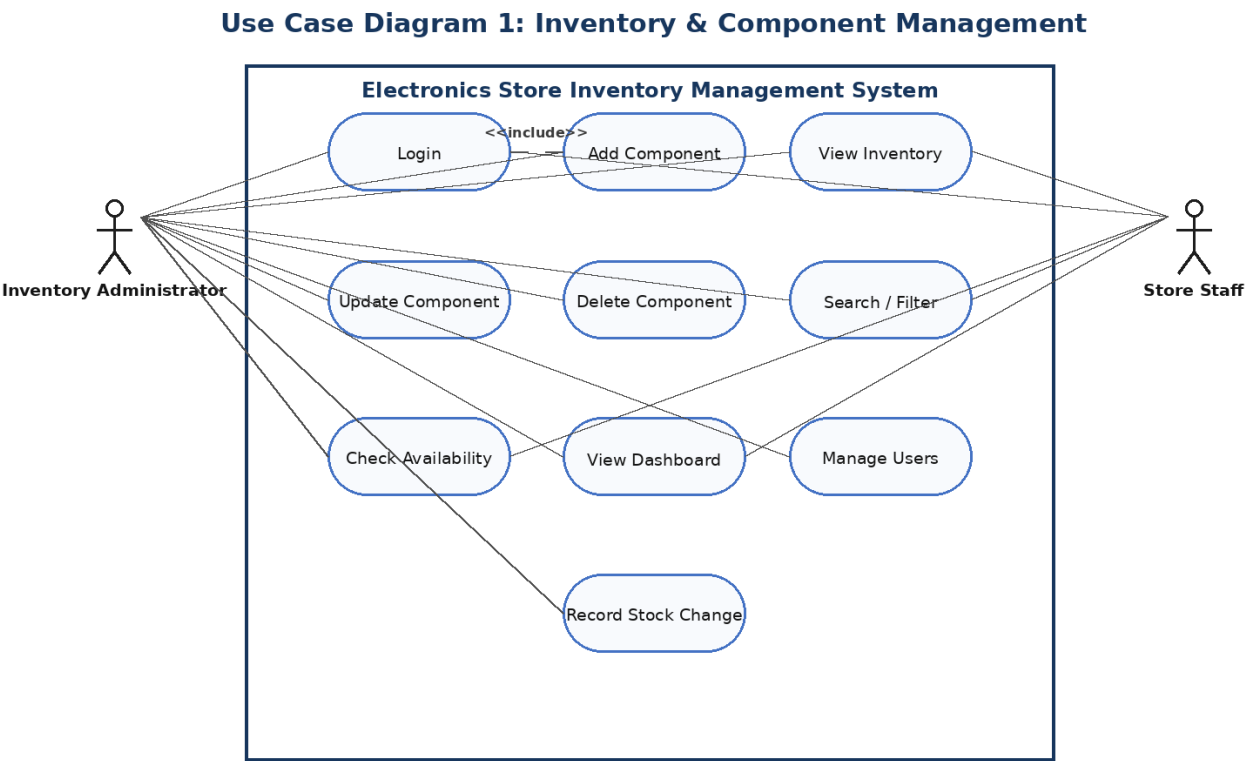


Figure 1 — Inventory and Component Management Use-Case Diagram

Diagram 2: Reporting, Availability & Monitoring

Actors: Store Manager and Inventory Administrator. Main use cases include login, dashboard, reports, availability, low-stock monitoring, report filtering/export, stock trends, reorder level and low-stock acknowledgement.

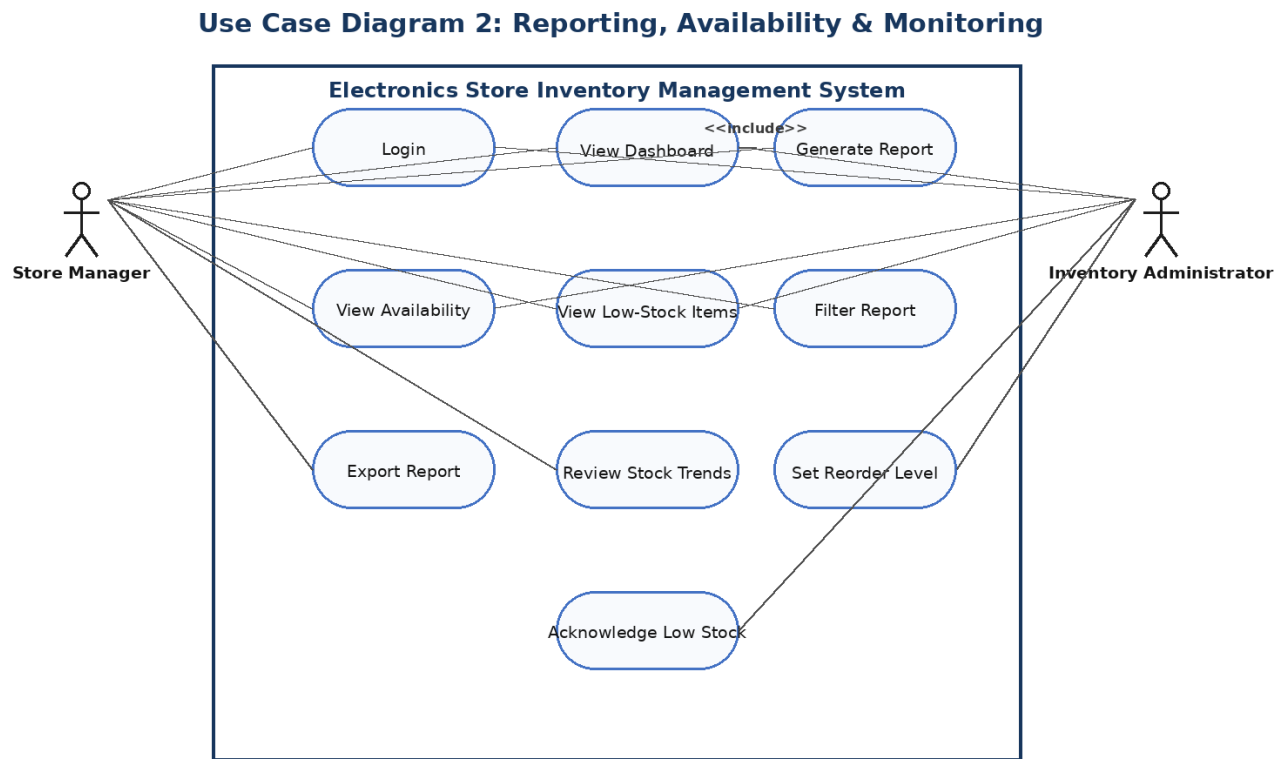


Figure 2 — Reporting, Availability and Monitoring Use-Case Diagram

7.3 Use-Case Diagram Alignment

The two original UML use-case diagrams are retained. The diagrams show the actor-to-use-case relationships for inventory/component management and reporting/availability/monitoring. Detailed supporting use cases such as filtering, exporting, stock trends, reorder level and low-stock acknowledgement are retained as part of the broader reporting, availability and monitoring workflow represented by the documented system features.

7.4 Design Specification Index

The table maps each design specification reference (DS-xx) from the RTM to its implementing module and the requirement it realizes.

Design Spec	Module	Requirement realized	SRS section
DS-01	AuthModule	ESIM-F-001 — User Authentication	4.1
DS-02	SessionManager	ESIM-F-002 — Logout & Session Management	4.2
DS-03	InventoryModule	ESIM-F-003 — Add Component	4.3
DS-04	InventoryView	ESIM-F-004 — View Inventory	4.4
DS-05	SearchFilterModule	ESIM-F-005 — Search / Filter	4.5
DS-06	InventoryModule	ESIM-F-006 — Update Component	4.6
DS-07	InventoryModule	ESIM-F-007 — Delete Component	4.7
DS-08	AvailabilityService	ESIM-F-008 — Availability Status	4.8
DS-09	StockAlertService	ESIM-F-009 — Low-Stock Identification	4.9

Design Spec	Module	Requirement realized	SRS section
DS-10	DashboardModule	ESIM-F-010 — Dashboard	4.10
DS-11	ReportModule	ESIM-F-011 — Inventory Reports	4.11
DS-12	ValidationModule	ESIM-F-012 — Input Validation	4.12
DS-13	Django ORM / Database	ESIM-F-013 — Persistent Storage	4.13
DS-14	AuthorizationModule	ESIM-F-014 — Authorization	4.14
DS-15	ErrorHandler	ESIM-F-015 — Error Handling	4.15
DS-16	AuditLogModule	ESIM-F-016 — Inventory Change Tracking	4.16
DS-NF-01	NFRModule	ESIM-NF-001 — Performance	5
DS-NF-02	NFRModule	ESIM-NF-002 — Scalability	5
DS-NF-03	NFRModule	ESIM-NF-003 — Reliability	5
DS-NF-04	NFRModule	ESIM-NF-004 — Availability	5
DS-NF-05	NFRModule	ESIM-NF-005 — Usability	5
DS-NF-06	NFRModule	ESIM-NF-006 — Maintainability	5
DS-NF-07	NFRModule	ESIM-NF-007 — Compatibility	5
DS-NF-08	NFRModule	ESIM-NF-008 — Recoverability	5
DS-SR-01	SecurityModule	ESIM-SR-001 — Security	5.1.2
DS-SR-02	SecurityModule	ESIM-SR-002 — Security	5.1.2
DS-SR-03	SecurityModule	ESIM-SR-003 — Security	5.1.2
DS-SR-04	SecurityModule	ESIM-SR-004 — Security	5.1.2
DS-SR-05	SecurityModule	ESIM-SR-005 — Security	5.1.2
DS-SR-06	SecurityModule	ESIM-SR-006 — Security	5.1.2